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Problem Set 4

Advanced Macroeconomics

Winter 2026/27

1 Solving for the Static Equilibrium in Dynare

1. Review Hands-on Exercise 3 from Chapter 3 of the Lecture.
2. Review Hands-on Exercise 5 from Chapter 3 of the Lecture.

2 Extension: One-Period Model with Public Sector

Consider the one-period model with endogenous labor from the lecture extended by consumption taxes and public spending. Assume the following utility function:

$$u(c,n) = \frac{c^{1-\sigma}-1}{1-\sigma} - b\frac{n^{1+\varphi}}{1+\varphi},$$

where b is a labor disutility weight. The public budget constraint is given as

$$\gamma y P \equiv gP = \tau c P,$$

where g is real public consumption, P is the price level, τ the consumption tax rate, and γ the share of public spending in GDP. The goods market clears if aggregate output is equal to aggregate demand:

$$y = g + c.$$

1. Derive the budget constraint and the first-order conditions for a utility maximizing representative household. Assume the same production technology and competitive firm behavior as in Chapter 3. All firm profits are distributed to the representative household.
2. Summarize the model: how many equations do we have? How many variables and parameters? Which are exogenous and which are endogenous?
3. Compute the general equilibrium using the toolbox Dynare.
4. Derive the static equilibrium analytically. Assume that one third of the available time is devoted to work $n = 1/3$. Calibrate b to obtain n=1/3.
5. Determine τ endogenously to achieve the public-spending target $\gamma = 0.20$.